MuseRAG++: A Deep Retrieval-Augmented Generation Framework for Semantic Interaction and Multi-Modal Reasoning in Virtual Museums

Yifan Hu

huyifan@ivt.edu.cn

Suzhou Industrial Park Institute of Vocational Technology

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Abstract

Virtual museums offer new opportunities for cultural heritage engagement by enabling interactive and personalized experiences beyond physical constraints. However, existing dialogue systems struggle to provide semantically adaptive, factually grounded, and multimodal interactions, often suffering from shallow user intent understanding, limited retrieval capabilities, and unverifiable response generation. To address these challenges, we propose MuseRAG++, a unified retrieval-augmented framework that integrates deep user intent modeling, a hybrid sparse-dense retrieval pipeline spanning text, images, and structured metadata, and a provenance-aware generation module that explicitly grounds responses in retrieved evidence. Unlike traditional methods relying on benchmark datasets, we evaluate MuseRAG + + through a qualitative user study within a functional virtual museum prototype, focusing on engagement, usability, and trustworthiness. Experimental results demonstrate substantial improvements in retrieval and generation metrics compared to strong baselines, while user evaluations confirm enhanced factual accuracy and interpretability.

1 Introduction

The digitization of cultural heritage [1] and the increasing accessibility of immersive technologies [2] have catalyzed the development of virtual museums, offering users a novel paradigm for interacting with historical artifacts and cultural narratives. Compared to traditional physical exhibitions, virtual museums offer asynchronous, location-independent access to curated content while also enabling novel forms of engagement through interactive interfaces [3]. As societal interest in cultural learning expands beyond institutional settings, there is a growing demand for intelligent systems that not only present digital replicas of artifacts but also support context-aware explanation, provenance reasoning, and personalized exploration. This positions the virtual museum as both a technological and educational frontier requiring deeper integration of human–AI interaction capabilities.

Prior research on virtual museums has explored a wide spectrum of interaction modalities and content presentation strategies. Early systems focused primarily on graphical reconstruction and spatial navigation [4], using 3D modeling [5], panoramic views [6], or VR-based walkthroughs [7] to simulate the physical visiting experience. More recent work has introduced rule-based or retrieval-based conversational agents [8], enabling users to query exhibit-related information through pre-scripted dialogues or keyword matching. Other approaches have leveraged recommender systems to personalize content delivery based on user profiles or browsing behavior [9]. Despite these advances, traditional virtual museum systems remain limited in semantic understanding and contextual adaptability. They often rely on static metadata or curated descriptions [10], which restricts their ability to accommodate open-domain queries or support in-depth, conversational engagement with cultural content.

With the advent of large language models (LLMs), virtual museum interaction has evolved toward more natural, adaptive, and knowledge-rich paradigms. LLMs, pre-trained on large-scale corpora encompassing both general and domain-specific knowledge, have demonstrated remarkable capabilities in open-domain question answering, contextual reasoning, and multi-turn dialogue generation [11–14]. These advances have catalyzed a new generation of virtual museum systems in which LLMs serve as intelligent cultural guides, enabling users to engage with exhibits through natural language interactions. Current LLM-based approaches to virtual museum dialogue typically adopt one of three architectural paradigms. The first relies on prompt-based generation [15], where instruction-tuned LLMs directly produce descriptions and responses based on user queries, offering flexibility but often lacking factual grounding. The second approach integrates retrieval-augmented generation [16], which augments language models with external evidence retrieval to enhance accuracy and contextual relevance. The third follows a modular or agent design [17], separating the interaction pipeline into discrete components for intent recognition, knowledge retrieval, and response generation, thereby improving interpretability and component-wise optimization. These architectures represent a significant advancement over rule-based or scripted systems, enabling more semantically enriched and conversationally fluid user experiences.

Several key challenges persist in effectively applying LLMs to cultural heritage interaction. A central limitation lies in the lack of robust mechanisms for modeling user intent, which is essential for generating responses aligned with nuanced informational needs, ranging from historical interpretation to stylistic analysis and provenance tracing. While retrieval-augmented methods have improved factual grounding, many implementations rely on shallow or unimodal retrieval strategies that fail to capture the rich heterogeneity of museum data, such as textual descriptions, visual representations, and structured metadata. Furthermore, current generation models often lack explicit provenance awareness, producing outputs that are fluent but difficult to verify. This lack of traceability undermines trust, particularly in cultural and educational contexts where accuracy and transparency of source are critical.

To address these challenges, we propose MuseRAG++, a retrieval-augmented interaction framework that enables semantically adaptive, verifiably grounded, and multi-modal dialogue in virtual museum environments. The framework introduces a unified architecture that integrates deep user intent recognition to infer contextual goals from multi-turn dialogue, a hybrid retrieval pipeline that combines sparse and dense semantic indexing to identify relevant evidence across text, images, and structured metadata, and a provenance-aware generation module that synthesizes coherent responses while explicitly tracing the source of each piece of information. By tightly coupling these components, our framework enables more precise user modeling, richer knowledge grounding, and transparent cultural interpretation, supporting human–AI interaction that is both adaptive and trustworthy.

Rather than relying on large-scale benchmark datasets, we evaluate MuseRAG + + through a user-centered qualitative study situated within a functioning virtual museum prototype. This methodological choice reflects the inherently experiential nature of cultural exploration, where standard accuracy metrics cannot fully capture user satisfaction, trust, and engagement. Through semi-structured interviews and interaction logging, we assess how different users perceive the relevance, credibility, and usability of the system. This approach provides richer insights into the system's real-world effectiveness and its capacity to support meaningful human–AI collaboration in cultural heritage settings. Our contributions are summarized as follows:

- We introduce MuseRAG++, a novel framework for virtual museum interaction that jointly models user intent, semantic retrieval, and provenance-guided response generation under a unified paradigm.
- We design a hybrid retrieval pipeline that combines BM25-based sparse retrieval with dual-encoder dense retrieval and LLM-guided semantic re-ranking, enabling fine-grained and multimodal grounding of cultural knowledge.
- We propose a provenance-aware generation mechanism that produces source-attributed responses, utilizing text, images, and metadata, thereby enhancing trust and transparency in artifact explanations.
- We conducted a qualitative user study using semi-structured interviews to assess engagement, usability, and perceived accuracy, demonstrating the practical value of our approach in real-world interactive museum scenarios.

2 Related Work

2.1 Virtual Museums

Virtual museums have emerged as key platforms for enhancing access to cultural heritage, offering digital representations of artifacts through spatially organized and visually immersive environments. Early systems focused on replicating physical exhibitions with the aid of 3D modeling [18], panoramic imaging [19], and interactive web interfaces [20]. As these techniques matured, many platforms incorporated virtual or augmented reality to simulate the spatial experience of traditional museum visits [21]. Beyond visual reconstruction, recent developments have emphasized content enrichment and user engagement. Features such as thematic browsing, multilingual annotations, interactive timelines, and recommendation systems have enabled more flexible and educationally oriented navigation [22]. Some large-scale platforms further support cross-collection search and metadata-based exploration, leveraging structured cultural databases to present artifacts in broader historical and stylistic contexts. In parallel, interaction modalities have evolved from static, point-and-click navigation toward more dynamic and adaptive forms [23]. Conversational interfaces, voice control, and guided tours have been introduced to support deeper engagement.

These developments have significantly enhanced accessibility and visual immersion, but current systems continue to face limitations in semantic understanding and interaction depth. Many systems continue to rely on rule-based logic or fixed dialogue flows, limiting their ability to interpret complex queries and address diverse informational needs. Key semantic and epistemic aspects of artifacts are often overlooked in user interfaces. Despite advances in visual fidelity, few platforms support dynamic synthesis of curatorial knowledge or mechanisms for tracing the origin and development of cultural objects. This gap reveals a persistent divide between the visual richness of virtual museums and the depth of semantic interaction users increasingly expect.

2.2 Large Language Models in Interactive Systems

LLMs have transformed the design of interactive systems by enabling more natural, adaptive, and semantically enriched human–machine communication [24]. Pretrained on large-scale corpora, LLMs possess strong capabilities in open-domain question answering, contextual reasoning, and dialogue generation [25–27]. These models have been increasingly adopted across various domains, including education [28] and healthcare [29], as well as cultural computing [30], where users often seek complex, context-dependent information beyond simple factual recall. In contrast to traditional rule-based or intent-matching dialogue systems, LLM-based approaches can interpret flexible user inputs, maintain discourse coherence, and generate informative responses across multi-turn interactions [31]. This has led to their use in a wide array of applications, from intelligent tutoring systems and museum guides [32] to virtual assistants for scientific and historical exploration [33]. Importantly, their ability to generate human-like responses has opened new opportunities for designing interfaces that simulate expert-like guidance and support open-ended inquiry.

To mitigate the limitations of pure generation, RAG frameworks have emerged as a compelling strategy [34]. These systems combine the reasoning abilities of LLMs with access to external knowledge sources to enhance factual grounding and reduce hallucination [35]. Complementary to this, modular pipelines decompose the interaction process into separate components for intent recognition, evidence retrieval, and controlled response generation, allowing for greater transparency and domain adaptability.

Despite these advances, several critical challenges remain in applying LLMs to complex, knowledge-intensive interactive settings. Many systems lack robust mechanisms for modeling user intent beyond surface-level semantics, making it difficult to tailor responses to different goals such as comparative analysis, historical tracing, or interpretive reflection. Furthermore, most retrieval components operate on unimodal textual corpora, underutilizing structured metadata and visual information that are often central to domains like cultural heritage. Finally, the absence of provenance tracking in most generative pipelines limits the credibility of responses, particularly in educational or archival contexts where source attribution is crucial. These limitations underscore the need for integrated frameworks that combine deep user intent modeling, multi-modal retrieval, and verifiable response generation, paving the way for more trustworthy and meaningful interactions in virtual museum environments.

3 Methods

3.1 Overview

1:Overview

MuseRAG + + is a unified framework designed for provenance-aware dialogue within virtual museum environments. It addresses the task of interactive cultural exploration, where users engage in multi-turn conversations to inquire about artifacts, styles, and historical contexts. Rather than treating the task as generic QA, MuseRAG + + explicitly models user intent, retrieves cross-modal evidence, and generates source-grounded responses.

The framework consists of three tightly integrated components. The user modeling module infers user intent based on conversational history. The hybrid retriever combines dense and sparse indexing to gather evidence from textual, visual, and structured sources, which is re-ranked for contextual relevance. The generation module synthesizes fluent and explainable responses, grounded in retrieved content.

3.2 User Intent Modeling

Effective dialogue grounding in virtual museum scenarios requires an accurate estimation of user intent that evolves with context. Instead of treating each query in isolation, the intent model in MuseRAG + + continuously updates a latent representation based on dialogue history and interaction signals. This representation guides downstream retrieval and generation by encoding semantic preferences, informational goals, and referential dependencies across turns.

Given a dialogue context $C_t = \{u_1, r_1, \ldots, u_{t-1}, r_{t-1}, u_t\}$, where u_i and r_i denote user and system utterances at turn i, we encode the context into a dynamic user representation $\varvec{z}_t$ through a transformer encoder:

$$\mathbf{z}_t = \mathrm{Encoder}\left(C_t\right). (1)$$

To disambiguate fine-grained user goals, we incorporate an auxiliary contrastive objective that distinguishes between semantically adjacent but pragmatically distinct intents. For each anchor query u_t, positive samples are generated from paraphrastic reformulations, while negatives are sampled from unrelated turns. The intent-aware embedding is refined using a supervised contrastive loss:

$$\mathscr{L}_{\text{intent}} = -log\frac{\exp\left(\frac{\mathrm{sim}\left(\mathbf{z}_{\mathrm{t}},\mathbf{z}_{\mathrm{t}}^{+}\right)}{\tau}\right)}{\sum_{\mathrm{j}}\exp\left(\frac{\mathrm{sim}\left(\mathbf{z}_{\mathrm{t}},\mathbf{z}_{\mathrm{j}}^{-}\right)}{\tau}\right)}, (2)$$

where $\mathrm{sim}\left(\bullet\right)$ denotes cosine similarity and τ is a temperature parameter. This contrastive refinement sharpens the semantic space of user representations, allowing intent-guided evidence retrieval to focus on provenance-relevant artifacts. As a result, the model supports long-horizon interactions where user interests shift over time and require contextual tracking and disambiguation.

3.3 Hybrid Semantic Retrieval

Effective information retrieval within virtual museum dialogues demands a balance between lexical precision and semantic generalization, while ensuring interpretability grounded in provenance.

MuseRAG++ addresses this by jointly encoding queries and candidate evidences into dual representations. A transformer-based encoder projects the user query q into a dense semantic embedding $\mathbf{q}_d \in \mathbb{R}^d$, capturing nuanced contextual and intent information. Simultaneously, a sparse vector $\mathbf{q}_s \in \mathbb{R}^{|V|}$ encodes term-level presence weighted by inverse document frequency, preserving critical exact-match signals. Candidate evidences $\mathbf{p}_i$ undergo analogous encoding, yielding $\mathbf{p}_{i,d}$ and $\mathbf{p}_{i,s}$, where the sparse component interfaces with a BM25-inspired scoring function.

Rather than aggregating these scores via static weighting, MuseRAG++ introduces an adaptive alignment function φ that contextualizes retrieval by evaluating the semantic congruence between candidate embeddings and the current dialogue state embedding \varvec$z_{\mathrm{t}}$. Formally, the retrieval score is defined as:

$$S(q, p_i) = \lambda \bullet < \mathbf{q}_d, \mathbf{p}_{i,d} > + (1 - \lambda) \bullet BM25(\mathbf{q}_s, \mathbf{p}_{i,s}) + \varphi(\mathbf{z}_t, \mathbf{p}_{i,d}), (3)$$

where $\lambda \in [0,1]$ governs the interpolation between dense and sparse components, while φ is a learnable mapping that encodes pragmatic relevance, refining retrieval outcomes beyond isolated lexical or semantic similarities.

In addition, a shared embedding space integrates heterogeneous modalities by means of a gating mechanism that dynamically weights textual $\mathbf{p}_{i,t}$, visual $\mathbf{p}_{i,v}$, and $\mathbf{p}_{i,m}$ representations:

$$\mathbf{p}_{i,d} = g_t \odot \mathbf{p}_{i,t} + g_v \odot \mathbf{p}_{i,v} + g_m \odot \mathbf{p}_{i,m}, g_t + g_v + g_m = 1, g_j = \sigma(\mathbf{W}_j [\mathbf{q}_d; \mathbf{z}_t]), (4)$$

where σ denotes the sigmoid activation and g_j the modality-specific gate values learned conditioned on both query semantics and dialog context. This retrieval framework is further refined via a joint optimization paradigm in which generation feedback informs the reranking parameters. This closed-loop learning strategy encourages the retrieval model to prioritize evidences that facilitate coherent, faithful response generation, effectively bridging the gap between retrieval fidelity and linguistic expression.

3.4 Provenance-aware Response Generation

To ensure that model-generated responses remain verifiable and faithful to retrieved evidence, we integrate a provenance-aware decoding mechanism guided by LLMs. This mechanism treats response generation as a constrained inference problem, where the output must be both contextually coherent and explicitly grounded in a structured set of supporting passages. Let $\mathscr{E} = \{e_1, e_2, \ldots, e_K\}$ denote the retrieved evidence set from the hybrid semantic retriever, and q be the user query.

Instead of naively concatenating $\mathscr{E}$ and prompting the LLMs with the full context, we first construct a provenance-aware latent plan $\pi = \{(s_t, e_t)\}_{t=1}^{T}$, where each step s_t in the response is aligned with an evidence fragment $e_t \in \mathscr{E}$, enforcing semantic traceability. To model this planning process, we maximize the joint likelihood:

$$\mathscr{L}_{\text{joint}} = \sum_{t=1}^{T} logP(s_t \mid s_{<t}, q, e_t) + logP(e_t \mid q, \mathscr{E}), (5)$$

where $P\left(e_t \mid q, \mathscr{E}\right)$ scores the salience of each evidence fragment given the query context, computed using a cross-attentive retriever-encoder; $P\left(s_t \mid s_{<t}, q, e_t\right)$ is implemented by the LLM through a knowledge-aware constrained decoding mechanism.

To prevent hallucinations and enforce provenance fidelity, we further regularize generation using a grounding loss that penalizes unsupported claims. For every output token y_t, we define a binary grounding indicator $g_t \in \{0,1\}$, and optimize:

$$\mathscr{L}_{\text{ground}} = -\sum_{t=1}^{T} g_t \bullet \, logP\left(y_t \mid y_{<t}, q, \mathscr{E}\right) + \left(1 - g_t\right) \bullet \, \log\left(1 - P\left(y_t \mid \bullet\right)\right). (6)$$

The final loss combines the planning and grounding objectives:

$$\mathscr{L} = \mathscr{L}_{\text{joint}} + \lambda \, \mathscr{L}_{\text{ground}}, (7)$$

where λ controls the trade-off between generation fluency and source traceability. Finally, the response is synthesized via an LLMs conditioned on both the structured evidence alignment plan and the original user query:

$$Response = \mathrm{LLMs}\left(q, \pi\,, \mathscr{E}\right). (8)$$

This process enforces a provenance-controllable generation paradigm, where each segment of the response is both semantically coherent and explicitly traceable to an evidence source. By embedding fine-grained alignment and uncertainty-aware decoding into the generative process, the model exhibits enhanced reliability and transparency, especially in high-stakes information tasks.

4 Experimentation

4.1 Dataset

To rigorously evaluate the effectiveness of the proposed provenance-aware response generation framework within a domain-specific setting, we construct a dedicated dataset centered on the Changxin Palace Lantern (□□□□), a prominent artifact in Chinese cultural heritage. This dataset is composed of a curated collection of 188 scholarly articles that encompass a wide range of perspectives, including archaeological interpretations, artistic analyses, and technological investigations pertaining to the lantern. In addition to textual sources, the dataset also incorporates five high-quality documentary videos, which offer rich multimodal descriptions and expert commentaries that further contextualize the historical and cultural significance of the object.

All data sources undergo systematic preprocessing to remove typographical noise, duplicated content, and irrelevant metadata. Textual segments are divided into coherent passages ranging from 100 to 300 words, ensuring semantic completeness and retrieval granularity. The resulting corpus is subsequently indexed to support structured and efficient evidence retrieval. This dataset serves not only as the external knowledge base during response generation but also provides the reference materials for grounding evaluation and provenance verification.

4.2 Implementation Details

Our framework is implemented using PyTorch with HuggingFace's transformers and accelerate libraries. For the retriever, we employ a hybrid strategy combining BM25 and a fine-tuned BGE-base embedding model, with top-10 passages selected per query. The response generator is based on Qwen3-8B [11]. For training the provenance-aware generation component, we use the AdamW optimizer with a learning rate of 2×10^{-5}, batch size of 4, and a maximum decoding length of 256 tokens. The λ coefficient is empirically set to 0.4.

4.3 Evaluation Metrics

To comprehensively assess the effectiveness of the proposed provenance-aware response generation framework, we employ a set of standard evaluation metrics that jointly measure both the retrieval quality and the factual integrity of the generated responses. Retrieval performance is quantified using Recall@k, which reflects the system's ability to successfully retrieve ground-truth supporting passages within the top-k candidates. Generation quality is primarily evaluated through precision, which indicates the proportion of statements in the generated responses that are factually accurate and grounded in the retrieved evidence. Furthermore, we compute accuracy to assess whether each complete response is semantically and factually consistent with the underlying knowledge, thereby capturing the holistic alignment between generation and provenance. To balance the trade-off between precision and recall, we additionally report the F1-score, representing the harmonic mean of these two metrics and providing an overall measure of response quality under a provenance-aware constraint.

4.4 Experimental Results

The proposed framework demonstrates performance across all evaluation metrics when compared with two competitive large language model baselines, as shown in Table 1. Specifically, our method achieves notable gains in Recall@10, Precision, Accuracy, and F1-score, indicating its ability not only to retrieve salient evidence but also to leverage that evidence effectively during generation. These improvements stem from the systematic integration of structured retrieval and provenance-constrained decoding, both of which contribute to factual alignment and response specificity.

The observed gains in recall are largely attributed to the hybrid retriever, which jointly optimizes semantic relevance and lexical salience. This design enables the retrieval module to capture both surface-level and conceptual cues within domain-specific corpora, particularly crucial in a setting like Changxin Palace Lantern where knowledge is sparse and terminology highly specialized. Moreover, the structured evidence planning mechanism facilitates the decomposition of generation into intermediate, evidence-aligned stages. This intermediate planning encourages more faithful content selection, mitigating the risk of drifting from grounded facts during decoding.

Table 1
Comparative Experimental Results

Models	Recall@10 ↑	Precision ↑	Accuracy ↑	F1 ↑
Qwen3-8b [11]	71.3	68.9	65.7	701
DeepSeek-R1-Distill-Qwen-7B [14]	73.5	70.2	68.0	71.8
MuseRAG++ (Ours)	**83.1**	**76.4**	**74.5**	**79.6**

Further performance gains arise from the grounding-aware decoding strategy, which injects explicit regularization against unsupported generations. This mechanism constrains the model to maintain coherence with selected evidence spans, thereby reducing hallucination and enhancing factual consistency. Notably, the increase in accuracy and precision reflects not only correct fact retrieval but also improved content selection at the generation level, indicating that the model does not merely copy retrieved passages but integrates them contextually and selectively.

The consistent performance advantages over Qwen3-8B [11] and DeepSeek-R1-Distill-Qwen-7B [14] suggest that generic large language models, even when fine-tuned, struggle to maintain provenance fidelity without structured retrieval and evidence alignment mechanisms. By contrast, the proposed approach bridges retrieval and generation through a provenance-aware interface, thereby aligning language modeling dynamics with factual grounding constraints. The consistent performance gains confirm that explicitly encoding provenance constraints into retrieval and decoding processes is instrumental in enhancing factual consistency and retrieval relevance, especially under the knowledge sparsity and semantic ambiguity inherent to cultural heritage scenarios.

4.5 User Study

To assess the human-perceived quality of generated responses, we conducted a user study involving 50 participants with backgrounds in both natural language processing and cultural heritage. The evaluation criteria are shown in Table 2. Each participant was presented with model outputs under blind conditions and asked to evaluate responses along 5 dimensions aligned with the central goals of provenance-aware generation. The results reveal a consistent advantage for our model across all metrics, particularly in factuality (4.5), evidentiality (4.6), and interpretability (4.7), suggesting that provenance-grounded modeling directly contributes to more reliable and transparent generation, as shown in Table 3.

Table 2
User Experience Evaluation Criteria.

Evaluation Dimension	Definition
Factuality	The response aligns with domain knowledge and supporting evidence.
Fluency	The language is natural, coherent, and grammatically correct.
Evidentiality	The generated facts are directly supported by retrieved passages.
Usefulness	The response adequately answers the query.
Interpretability	The user can trace claims back to cited passages.

Table 3
Subjective Evaluation Scores.

Models	Factuality	Fluency	Evidentiality	Usefulness	Interpretability	Avg
Qwen3-8b [11]	3.7	4.1	3.3	3.6	3.8	3.7
DeepSeek-R1-Distill-Qwen-7B [14]	3.9	4.2	3.6	3.8	4.2	3.9
MuseRAG++ (Ours)	4.5	4.4	4.6	4.5	4.7	4.5

Compared to the baseline models, which exhibit moderate performance in factual alignment and source traceability, our approach demonstrates a sharper ability to preserve semantic alignment between retrieved passages and the generated content. The elevated interpretability score underscores the model's capacity to expose its reasoning path through explicit grounding in retrieved materials, addressing a critical requirement in domains where epistemic trust and accountability are essential. Moreover, while baseline systems maintain a degree of fluency, they often fail to sustain evidential integrity, leading to plausible yet unverifiable statements. Our method mitigates this tendency by incorporating provenance signals throughout the decoding process, which not only reinforces factual consistency but also improves user confidence in the model's claims.

To quantify the reliability of subjective evaluations, we calculated the intraclass correlation coefficient (ICC) using a two-way mixed-effects model ICC(3,k) across 50 raters. The resulting ICC was 0.962, with a 95% confidence interval of [0.941, 0.978], indicating excellent inter-rater consistency across five evaluation dimensions. The associated F-statistic of 25.91 ($df_1 = 4$, $df_2 = 196$, $p < 0.0001$) further validates the robustness of scoring agreement, confirming that the quality gains achieved by MuseRAG + + are consistently perceived.

These results suggest that the integration of retrieval-grounded architectural design and evidence-aware decoding does not merely refine linguistic quality, but also structurally reorients generation toward verifiable, context-sensitive, and user-aligned responses.

5 Conclusion

In this paper, we propose MuseRAG++, a provenance-aware retrieval-augmented framework that facilitates semantically aligned, evidence-grounded, and multimodally adaptive dialogue in virtual museum environments. By jointly modeling user intent, hybrid retrieval across sparse-dense semantic spaces, and source-attributed generation, MuseRAG + + addresses critical challenges in factual grounding and interpretability in cultural dialogue systems. The unified architecture enables dynamic user modeling and multimodal knowledge grounding while maintaining transparency through explicit provenance tracing. Rather than relying solely on quantitative benchmarks, we adopt a user-centered qualitative evaluation within a functional museum prototype, revealing high levels of perceived relevance, credibility, and engagement, with strong inter-rater consistency. These results demonstrate the system's effectiveness in supporting trustworthy and context-aware human–AI interaction in cultural heritage settings. This work underscores the importance of provenance modeling and interaction-grounded evaluation in the design of next-generation dialogue agents for interpretive and experiential domains.

Declarations

Author Contribution

Y conducted all experiments, analysed the data, prepared all figures and tables, and wrote the manuscript. Y reviewed and approved the final version of the paper.

Data Availability

The questionnaire data and database information are available at:https://github.com/Alkaid-Hu/MuseRAG-The code will also be published at this address after the article is published.

References

1. Buragohain D, et al. Digitalizing cultural heritage through metaverse applications: challenges, opportunities, and strategies. Herit Sci. 2024;12(1):295.
2. Wang H et al. Grand challenges in immersive technologies for cultural heritage. Int J Human–Computer Interact (2025): 1–22.
3. Lucchi E. Digital twins, artificial intelligence and immersive technologies for heritage preservation and cultural tourism. smart cities. Digital Twin, Blockchain, and Sensor Networks in the Healthy and Mobile City. Elsevier; 2025. pp. 507–20.
4. Zhang Y, et al. Enhancing spatial cognition in online virtual museum environments: integrating game-based navigation strategies for improved user experience. Appl Sci. 2024;14(10):4163.
5. Spallone R, et al. 3D modelling and virtual reality for museum heritage presentation: Contextualisation of sculpture from the tang era. Int Archives Photogrammetry Remote Sens Spat

Inform Sci. 2024;48:413–20.

6. Hu Q, et al. Hybrid three-dimensional representation based on panoramic images and three-dimensional models for a virtual museum: Data collection, model, and visualization. Inform Visualization. 2017;16(2):126–38.
7. Zidianakis E et al. The invisible museum: A user-centric platform for creating virtual 3D exhibitions with VR support. Electronics 10.3 (2021): 363.
8. Krataithong P et al. Tourism Chatbot Framework: Enhancing Visitor Experience Through GraphRAG and AI Chatbot. 2025 IEEE International Conference on Cybernetics and Innovations (ICCI). IEEE, 2025.
9. Rodriguez-Boerwinkle RM, Paul J. Silvia. Visiting virtual museums: How personality and art-related individual differences shape visitor behavior in an online virtual gallery. Empir Stud Arts. 2024;42(2):439–68.
10. Zhang S et al. The Effects of LLM-Empowered Chatbots and Avatar Guides on the Engagement, Experience, and Learning in Virtual Museums. Int J Human–Computer Interact (2025): 1–13.
11. Yang A et al. Qwen3 technical report. arXiv preprint arXiv:2505.09388 (2025).
12. Hurst A et al. Gpt-4o system card. arXiv preprint arXiv:2410.21276 (2024).
13. Grattafiori A et al. The llama 3 herd of models. arXiv preprint arXiv:2407.21783 (2024).
14. Guo D et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. arXiv preprint arXiv:2501.12948 (2025).
15. Ly C et al. Museum in the Classroom: Engaging Students with Augmented Reality Museum Artifacts and Generative AI. Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems. 2025.
16. Masasi de Oliveira E, Ayumi et al. Immersive Virtual Museums with Spatially-Aware Retrieval-Augmented Generation. Proceedings of the 2025 ACM International Conference on Interactive Media Experiences. 2025.
17. Schott E et al. The Guide's Apprentices: Engaging Visitors of Virtual Museums through Appropriate Agent Embodiments. Proceedings of the 30th ACM Symposium on Virtual Reality Software and Technology. 2024.
18. Zhao G, Thienmongkol R, and Ruethai Nimnoi. Application of Image Recognition and 3D Reconstruction Technology in Virtual Museum System. 2024 International Conference on Advances in Electrical Engineering and, Applications C. (AEECA). IEEE, 2024.
19. Yang J. Design of Virtual Roaming System of Art Museum based on VR Technology. Scalable Computing: Pract Experience. 2025;26(3):1466–73.
20. Li Y et al. Human-Centric Virtual Museum: Redefining the Museum Experience Through Immersive and Interactive Environments. Int J Human–Computer Interact (2024): 1–12.
21. Tian F, et al. An empirical study of virtual museum based on dual-mode mixed visualization: the Sanxingdui bronzes. Herit Sci. 2024;12(1):146.

22. Ai L, Phaholthep C. Evaluating museum environment composition containing digital media interaction to improve communication efficiency. Buildings. 2025;15(7):1186.
23. Kim H, et al. Space-adaptive artwork placement based on content similarities for curating thematic spaces in a virtual museum. ACM J Comput Cult Herit. 2024;17(1):1–21.
24. Borghoff UM, Bottoni P, Pareschi R. Human-artificial interaction in the age of agentic AI: a system-theoretical approach. Front Hum Dynamics. 2025;7:1579166.
25. Shailendra P et al. Survey of Large Language Models for Answering Questions Across Various Fields. 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS). Vol. 1. IEEE, 2024.
26. Lee J, Yang K-A, Lee, Woon-Seng G. Modeling the One-to-Many Property in Open-Domain Dialogue with LLMs. arXiv preprint arXiv:2506.15131 (2025).
27. Yu J et al. LLMs as Collaborator: Demands-Guided Collaborative Retrieval-Augmented Generation for Commonsense Knowledge-Grounded Open-Domain Dialogue Systems. Findings of the Association for Computational Linguistics: EMNLP. 2024. 2024.
28. Neumann A, Tobias et al. An llm-driven chatbot in higher education for databases and information systems. IEEE Trans Educ (2024).
29. Wang Z et al. Healthq: Unveiling questioning capabilities of llm chains in healthcare conversations. Smart Health (2025): 100570.
30. Li C, et al. Culturellm: Incorporating cultural differences into large language models. Adv Neural Inf Process Syst. 2024;37:84799–838.
31. He R et al. Research and Implementation of Task-oriented Dialogue Systems Based on Large Language Models and Vector Retrieval. 2024 4th International Conference on Electronic Information Engineering and Computer Communication (EIECC). IEEE, 2024.
32. Vasic I et al. Llm-aided museum guide: Personalized tours based on user preferences. International Conference on Extended Reality. Cham: Springer Nature Switzerland, 2024.
33. Ren S et al. Towards scientific intelligence: A survey of llm-based scientific agents. arXiv preprint arXiv:2503.24047 (2025).
34. Singh A et al. Agentic Retrieval-Augmented Generation: A Survey on Agentic RAG. arXiv preprint arXiv:2501.09136 (2025).
35. Chen X et al. Honest AI: Fine-Tuning Small Language Models to Say I Don't Know, and Reducing Hallucination in RAG. arXiv preprint arXiv:2410.09699 (2024).

Figures

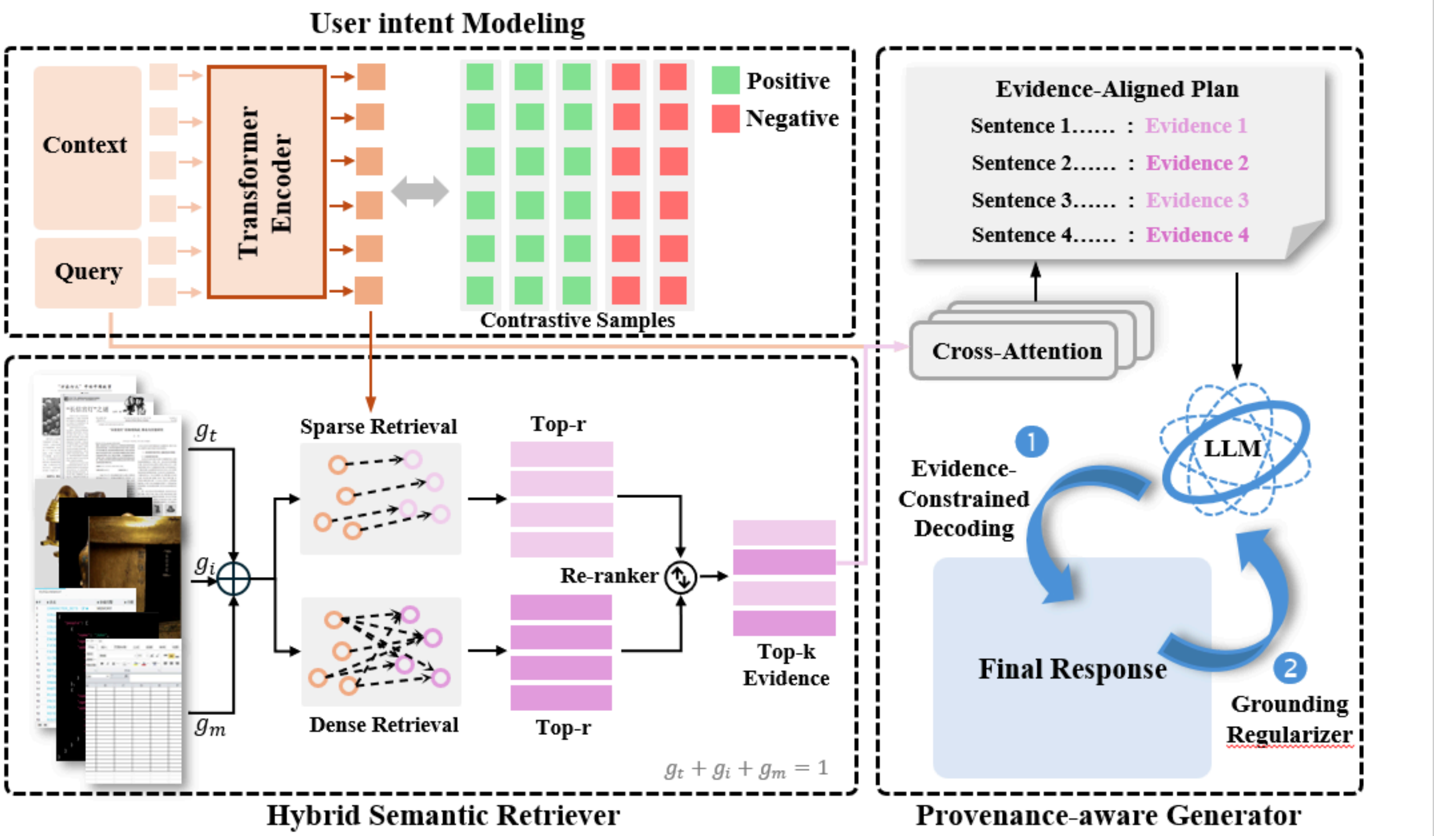

Figure 1

Unnumbered image in the Methods section.